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Cost-effective sensor placement optimization for large-scale urban sewage surveillance

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ABSTRACT

Early pandemic outbreak detection in cities is a crucial but challenging task. Complementary to the costly massive individual testing, urban sewage surveillance offers a rare, cost-effective solution for large-scale monitoring of pandemic spread in cities with minimal interference to people's lives. One emerging question is how to derive a cost-effective sensor placement plan in city-scale sewage networks having complicated topologies. Inspired by remote sensing, we first provide a general multi-objective formulation of the optimal sensor placement problem on directed networks. Then, we introduce a connectivity-based objective evaluation approach and embed it into an NSGA-II algorithm to enable efficient optimization on large-scale directed graphs. The effectiveness of the proposed method is verified on a real-world sewage network in Hong Kong serving more than 500,000 urban residents. Results show that the proposed method efficiently generated optimal sensor placement plans on city-scale networks. Optimized sensor placement plans outperformed human placement heuristics by a significant margin of 102%, highlighting the necessity for data-driven decision support for large-scale urban sensing. Methodologically, this study provides a benchmark problem and datasets for network-based spatial optimization studies. Codes and datasets developed in this study are open-sourced to support future research in a real-world scenario.

1. Introduction

Early pandemic outbreak detection is imperative in containing the spread of viruses within communities, especially in the case of highly contagious viruses with a high reproductive number (R_0) (Moosazadeh, Ifaei, Charmchi, Asadi, & Yoo, 2022). Therefore, over 55 countries worldwide have adopted sewage surveillance to detect the outbreak and monitor the spread of COVID-19 in communities (Ahmed et al., 2020; Arora et al., 2020; Deng et al., 2022; Gude & Muire, 2021; Naughton et al., 2023), highlighting the unique advantages offered by sewage-based underground urban sensing (Asif et al., 2022). Unlike massive individual viral testing, sewage surveillance collects and analyzes water samples from underground sewage networks without interfering with the daily lives of citizens (Mercer & Salit, 2021). Besides, each sewage water sample reflects the collective health conditions of the entire upstream population, making sewage surveillance a cost-efficient approach for city-scale urban health surveillance. Urban sensing is an emerging field for collecting information from urban spaces to understand urban phenomena and dynamics, yet insufficient attention has been paid to urban sensing from the underground. A recent MIT report (Hamzelou, 2023) highlights the prospect of city-scale sewage surveillance systems as an urban sustainable infrastructure to enable long-term, post-pandemic urban health monitoring, such as antibiotic-resistant virus monitoring (Aarestrup & Woolhouse, 2020; Larsson, Flach, & Laxminarayan, 2022) and community drug use detection (Reid, Baz-Lomba, Ryu, & Thomas, 2014; Yang, Castrignanò, Estrela, Frost, & Kasprzyk-Hordern, 2016).

However, designing a cost-effective sewage surveillance system for a city is a challenging task. On the one hand, a number of sensors need to be placed in a large-scale network to achieve the desired performances collectively. The computational complexity increases dramatically with the network size and number of sensors. On the other hand, cities have complicated sewage networks involving many nodes and links. Such topological complexity goes beyond human logical capabilities to identify an optimal sensor placement plan. Moreover, cost-effective sewage surveillance involves multiple objectives in natural conflict, for example, the trade-off between coverage and sensing budget. Therefore, attempts have been made to formulate and solve the optimal sensor placement (OSP) problem using optimization methods (Yang

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et al., 2016). The OSP problem has received sustained attention in various application fields, such as air pollution monitoring (Mulligan & Ammari, 2010; Wang, 2010), wireless anchor positioning (Vecchio, López-Valcarce, & Marcelloni, 2012), and water contamination detection (Adedoja, Hamam, Khalaf, & Sadiku, 2018b; Ciaponi et al., 2019).

In an OSP, a group of sensors gets placed in a bounded solution space to achieve desired objectives and satisfy feasibility constraints collectively. Many OSP practices have been developed on water distribution networks for water quality monitoring or leakage detection (Ciaponi et al., 2019; Krause, Leskovec, Guestrin, VanBriesen, & Faloutsos, 2008; Xing, Raviv, & Sela, 2022), yet challenges remain for OSP on sewage networks to enable city-scale sewage surveillance (Calle et al., 2021; Larson, Berman, & Nourinejad, 2020; Wang, 2010). First, most existing literature used single-objective formulations, such as maximizing coverage (Calle et al., 2021), minimizing costs (Nourinejad, Berman, & Larson, 2021), and balancing the effective coverages among sampling sites (Kim et al., 2022), while sewage surveillance naturally involves multiple desired objectives that conflict with each other. Second, insufficient attention has been paid to the directedness of networks and upstream-downstream relationships, an important property of sewage networks. Third, most OSP algorithms were only tested on small-scale synthetic networks (Larson et al., 2020), while optimization algorithms need to be efficient and scalable for city-scale sewage networks including a large number of nodes and links. The OSP problem involves computationally expensive and spatially explicit objectives. Leskovec et al. (2007) proposed a lazy greedy algorithm (CELF) to scale OSP to large-scale networks based on the observation that objectives in the OSP problem mostly share the submodularity property. However, the proposed approach adopts a single-objective formulation and only focuses on the submodularity property of objectives, while the increased complexity in large-scale OSP problems mainly originates from the significantly increased size of city-scale sewage networks. Insufficient attention has been paid to network properties and simplifications for the efficient generation of OSP solutions.

In this paper, we revisit the OSP problem and propose an improved model for cost-effective sensor placement to enable sewage surveillance at the city scale. First, we propose a multi-objective formulation for cost-effective sewage surveillance. Then, we solve this problem at a city scale using an NSGA-II algorithm coupled with an efficient network-based objective evaluator. We demonstrate the effectiveness of the proposed model using a real-world case study in Hong Kong and compare model performances with human placement heuristics.

The remainder of this paper is organized as follows. Section 2 briefly reviews relevant work. Section 3 presents methods, including model formulation, optimization algorithms, sewage network dataset, and performance measurement. Results are reported in Section 4 and discussed in Section 5. Section 6 concludes.

2. Related work

The optimal sensor placement (OSP) problem, which originated in the 1940s, has been applied in various fields of practice, such as air pollution monitoring (Mulligan & Ammari, 2010; Wang, 2010), wireless anchor positioning (Vecchio et al., 2012), and water contamination detection (Adedoja et al., 2018b; Ciaponi et al., 2019; Ogie, Shukla, Sedlar, & Holderness, 2017). Among these applications, contamination detection in sewage networks has received significant attention during the Covid-19 pandemic (Larson et al., 2020; Nourinejad et al., 2021; Wang & Liu, 2021) due to its practical benefits in disease control. However, to the best of our knowledge, the identification of an optimal sensor placement strategy for sewage surveillance was not available prior to the work of Larson et al. (2020), who framed this problem and proposed a Bayesian probability-based binary search to find the location of patient zero and tested their algorithm in the Marlborough sewage network. However, their algorithms are limited to the network with one wastewater treatment plant. Wang et al. (2020) developed a simulation model for pathogen shedding, transport, fate, sampling and detection in the sewage network and put forward an adaptive sampling site allocation method to maximize the probability of detecting the virus based on their performance. They carried out a case study based on the sewage system in Kolkata, which has 1000 edges and the upstream and downstream relationship is assumed unknown. Calle et al. (2021) proposed a greedy-based monitoring site selection algorithm to select large, non-interference, equal-sized coverage areas for static monitoring. Furthermore, they extended the method proposed by Larson et al. by assigning Bayesian probabilities based on the information of inhabitants for dynamic monitoring. The case study in Girona shows that 5–7 monitoring stations can achieve 80% coverage though the number of manholes monitored per station is large. Kim et al. (2022) proposed a spatial bisection method (SBM), which is extended from Larson et al. (2020) and genetic algorithm (GA) which aims to minimize the maximum effective monitoring area across monitoring stations. They carried out a case study in Suwon City, Republic of Korea by selecting 699 manholes from the sewage network with 5424 manholes and found that GA is less effective in evenly allocating monitoring areas to manholes compared to SBM, which allows stepwise selection of sampling manholes.

The optimization of sensor placement has received significant attention in the context of water distribution networks (Adedoja, Hamam, Khalaf, & Sadiku, 2018a; Giudicianni et al., 2020), while there is limited research on optimal sensor placement for sewage surveillance (Calle et al., 2021; Mac Mahon, Criado Monleon, Gill, O'Sullivan, & Meijer, 2022). However, these problems exhibit a shared underlying structure: given a network, identify a subset of nodes that can effectively detect the occurrence of a signal while minimizing costs and maximizing efficiency (Leskovec et al., 2007). The common objectives of the OSP are usually related to quick virus identification and minimizing the public health risk (Adedoja et al., 2018a). Maximizing the coverage is a major concern in OSP because the sewage network can be assumed guaranteed if a large proportion of the network is monitored (Adedoja et al., 2018a) and it has been included as one of the objectives by many researchers (Kumar, Kansal, & Arora, 1997; Lee & Deininger, 1992; Liu, Liu, Chen, & Wang, 2012). However, only considering coverage is not sufficient. Selecting sensors far from the contamination event to cover more nodes may cause inefficient detection and expose more people to the virus. Therefore, contamination detection is a multi-objective optimization problem in nature and some additional objectives are proposed for a more reasonable formulation, such as improving the robustness of the sensor system (Xing et al., 2022), increasing the detection efficiency of the configuration (Lalou, Tahraoui, & Kheddouci, 2018), reducing detection time, the number of sensors (Krause et al., 2008), and population exposed to the virus (Berry, Hart, Phillips, Uber, & Watson, 2006) etc.

Various optimization algorithms have been developed and implemented to solve the OSP problem, including both exact methods (Janke et al., 2012; Marchese, Jin, Fox-Lent, & Linkov, 2020) and heuristic methods (Candelieri, Ponti, & Archetti, 2021; Wettergren & Costa, 2021). Propato (2006) formulated a mixed-integer linear program and solved it using branch and bound techniques. Berry et al. (2006) proposed a MIP formulation to track contamination events and solved it using ILOG's AMPL/CPLEX 9.1 MIP solver. They further extended it to model imperfect sensors (Berry et al., 2009). However, due to the high memory consumption and complexity of the problem (Krause et al., 2008), the MIP approach cannot find the optimal solution in polynomial time and is not scalable to large-scale networks. Therefore, researchers have begun to explore heuristic algorithms. For example, Krause et al. (2008) proposed an algorithm based on simulated annealing and greedy algorithm and applied it to a network with 21,000 nodes. They also employed local search methods in the constraint generation process. The algorithm generates solutions within 98% of optimality with 1/100 of the run time of traditional MIP approaches.

The mentioned approaches are primarily designed for undirected graphs, whereas many real-life graphs are directed. Most existing methods either rely on the dynamic modeling of hydraulic information (Hart & Halden, 2020) or the dynamic spreading process of the contamination (Ciaponi et al., 2019; van Lieverloo, Mesman, Bakker, Baggelaar, Hamed, & Medema, 2007), which are computationally expensive and data-intensive. However, solving the OSP problem for sewage surveillance is different because the flow in the sewage network is unidirectional, making the simplification and idealization of network configuration possible (Kim et al., 2022). Treating directed graphs as undirected graphs results in a loss of directional information, limiting their ability to capture explicit sensor topological relationships, such as upstream-downstream connections. The key idea in sewage surveillance is that the manholes can be considered safe once the downstream sensors report no virus signal (Larson et al., 2020; Nourinejad et al., 2021). As the number of sensors increases, the upstream-downstream relationship between sensors becomes complicated and needs to be processed carefully to calculate the monitoring area of a sensor since there exists overlapping. Kim et al. (2022) adopted a squared binary matrix to store the connectivity information of all manholes and monitoring areas recursively. Similarly, in the water distribution network domain, Lee and Deininger (1992) generated a covering matrix to represent the covering ability of each sensor. Some other researchers adopted this idea and matrix (Kessler, Ostfeld, & Sinai, 1998; Ostfeld & Salomons, 2004), yet the matrix is sparse and computationally expensive, leading to inefficient solutions and limited applicability.

3. Methodology

3.1. Problem formulation

Given a sewage network with M manholes, let $X = (x_1, x_2, \ldots, x_M)$ denote a sensor placement plan, where $x_i = 1$ denotes a sensor placed at manhole i, and $x_i = 0$ denotes a manhole without a sensor. Then, the multi-objective sensor placement (MOSP) model aims to generate an optimal set of sensor placement plans $X_1, X_2, \ldots, X_I$ for cost-effective sewage surveillance by balancing the trade-off among the objectives.

In this study, we formulate the sensor placement problem following a cost-coverage-resolution (CCR) objective framework inspired by remote sensing tasks (Marceau & Hay, 1999; Woodcock & Strahler, 1987). Each sensor placement plan is evaluated for three objectives: (1) minimizing sensing cost, (2) maximizing sensing coverage, and (3) minimizing the expected search cost. While Objective 1 minimizes the sensing budget, Objective 2 maximizes the effective coverage of a sensor placement plan. Objective 3 is inspired by the concept of spatial resolution of satellite imagery. In a sensing task, the observational resolution measures the level of observed details provided by a sensing plan. It comes with a trade-off with the coverage and sensing cost.

Sensing cost. The sensing cost, N, denotes the cost of a sensor placement plan, measured as the total number of sensors placed in a sensor placement plan (Eq. (1)). We adopt a unit-cost assumption in which the placement cost of each sensor, c_i is uniform, while the formulation can be extended to situations where c_i varies at different locations. In practice, the total sensing cost is usually bounded by a budget.

$$N = \sum_{i=1}^{M} c_i x_i \tag{1}$$

Sensing coverage. The sensing coverage, C, measures the effective coverage of a sensor placement plan, calculated as the total number of manholes covered by sensors in a sensor placement plan in the entire sewage network. Let C_i denote the coverage of a sensor i, calculated as,

$$C_i = \sum_{j=1}^{n_i} p_j \tag{2}$$

where n_i is the number of manholes covered by sensor i, and p_j denotes a priority score of manhole j that is covered by sensor i. The priority score could either be measured as the population size or pandemic risk factor associated with manhole j. For example, a manhole connected to an elderly house could have a larger p_j value than other manholes, so this manhole will be prioritized in the MOSP model. This study adopts a uniform priority scenario for computational experiments, yet the methodology can be extended to a prioritized scenario. Then, the total coverage of a set of sensors can be calculated as:

$$C = \sum_{i=1}^{N} C_i O_i \tag{3}$$

where $O_i = 1$ if there is no sensor placed in its downstream area, else $O_i = 0$. We introduce O_i to capture the upstream-downstream relationship of sensors since there are overlaps between the upstream areas of different sensors and the total coverage of a set of sensors does not equal the sum of the coverage of all sensors. An example can be found in Fig. 1.

Expected search cost. The expected search cost, S, measures the spatial resolution of a sensor placement plan as the expected number of manual tests required to identify the exact location of a viral outbreak. In the extreme case where every manhole is sensor-placed ($N = M$), the expected search cost is 0 since any positive signal can be traced directly to a location. In a practical scenario where $N \ll M$, the sensor placement plan first narrows the target area down to a branch of the sewage network having n_i manholes. Then, additional sewage sampling steps are required to further identify the viral outbreak location. Multiple methods have been proposed to measure the expected search cost on a tree network (Larson et al., 2020; Nourinejad et al., 2021).

We adopt a formulation based on Bayesian probability (Nourinejad et al., 2021) to measure the expected search cost, S_i, on a network branch having n_i manholes and a single origin exclusively covered by a sensor i (Eq. (4)). The calculated expectation corresponds to the worst case in a binary search. Detailed proof can be found in Nourinejad et al. (2021). For large-scale networks, the expected search cost for the entire network can be aggregated using relative coverage weights (Eq. (5)). Fig. 1 demonstrates the expected search cost calculation on an example network containing three sensors, A, B, and C. First, each sensor's additional search cost is calculated respectively using Eq. (4) and their exclusive coverage is recorded. Then the expected search cost is calculated using Eq. (5) and equals 1.92.

$$S_i = log_2(n_i) \tag{4}$$

$$S = \sum_{i=1}^{N} \frac{n_i}{C} \cdot S_i \tag{5}$$

3.2. Efficient objective evaluation on large-scale directed networks

We propose an efficient connectivity-based objective evaluation method to enable optimal sensor placement on large-scale sewage networks (Fig. 2). First, a connectivity sub-graph, G', gets extracted for a given sensor placement solution by measuring whether the sensor-placed manholes are connected in the original graph. If two sensors are connected in the original graph, an edge is added to connect them in the connectivity subgraph. If no connection exists, the sensors are retained as isolated nodes in the sub-graph (Fig. 2a–d). Then, objective values are calculated using the connectivity subgraph. Since the size of the connectivity subgraph is significantly smaller than the original graph, objective evaluations based on the connectivity subgraph can be significantly efficient and scalable.

We further adopt a backward search strategy when the upstream of multiple sensor nodes overlap. First, we recognize the upstream-most sensors in each interconnected network component and calculate the additional search cost for these sensors. Then, we remove them from

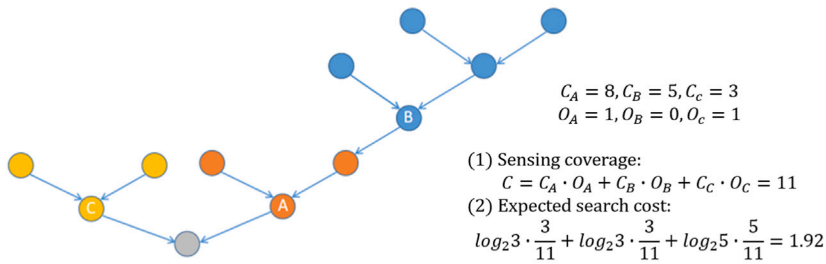


Fig. 1. The calculation of (1) sensing coverage and (2) expected search cost on a network example.

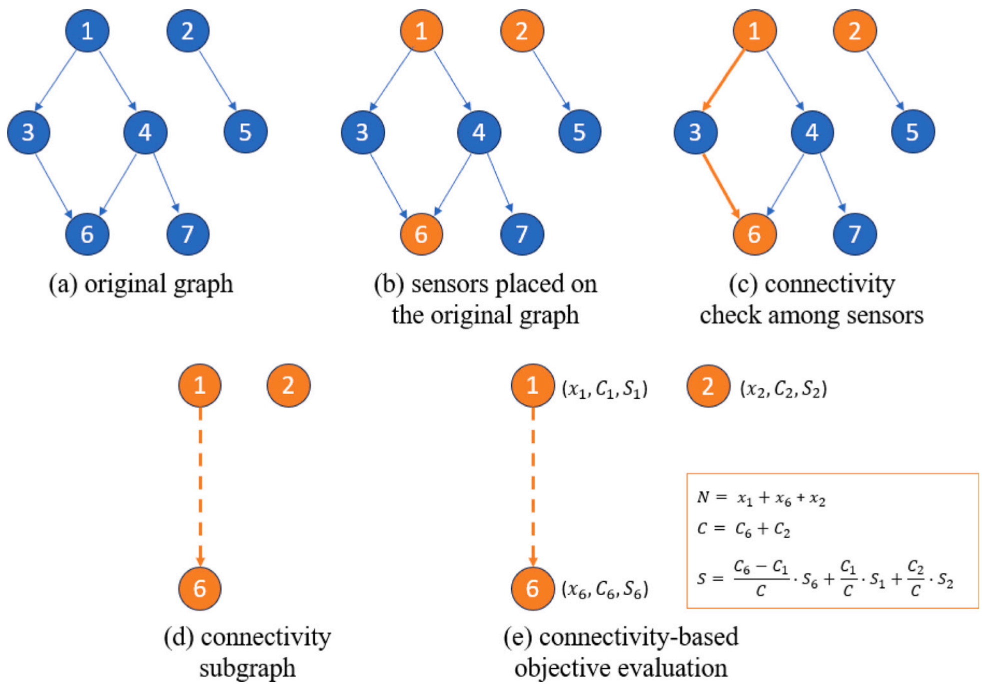


Fig. 2. The conceptual flow of (1) connectivity subgraph extraction (a–d) and (2) connectivity-based objective evaluation (e). In the original graph (a), sensors 1 and 6 are connected through path 1 → 3 → 6 while sensor 2 is independent. Therefore in the connectivity subgraph (d), all sensors remain as nodes and edge 1 → 6 is added to represent the upstream-downstream relationship among sensors.

the sub-graph and move on to the most upstream sensors in the updated subgraph until it is empty. After that, we aggregate the additional search cost of each sensor by weighting them based on the number of manholes they cover exclusively. Please refer to the Appendix for detailed algorithm steps.

3.3. Multi-objective optimization algorithms

According to Sela and Amin (2018), there are four types of approaches for the optimal sensor placement problems: (1) mixed-integer optimization, (2) greedy approximations, (3) evolutionary algorithms, and (4) human placement heuristics. In this study, we use evolutionary algorithms to solve the MOSP model. Optimized results are compared with human heuristics based on complex network topologies.

Non-dominated Sorting Genetic Algorithm (NSGA-II). We implement a standard NSGA-II algorithm (Deb, Agrawal, Pratap, & Meyarivan, 2002) to solve the OSP problem (Fig. 3). First, a number of initial sensor placement plans get initialized for a given sewage network, forming an initial population for the evolution process. Objective values of all plans are calculated using the network-based objective evaluator. Then, elitist sorting among plans in the population is performed using a non-dominated sorting algorithm (Deb et al., 2002). When comparing plan X_0 with plan X_1, X_0 dominates X_1 if and only if (1) X_0 is superior to X_1 in at least one of the objectives, and (2) X_0 is at least as good as X_1 in all other objectives. When sorting many plans in a population, each plan is compared with every other plan to figure out pairwise domination relationships. Then, plans not dominated by any other plan are placed on the first front. Plans only dominated by first-front solutions are placed on the second front. The process goes on until all plans have been placed on fronts. To further differentiate plans within a front, a crowding distance metric is calculated to promote diversity among plans. After elitist sorting, the better half of the population goes through the evolution process to generate an equal number of offspring, while the other half of the population gets eliminated. A typical evolution process includes a crossover operation and a mutation operation. We adopt a single-point crossover operator and a bit string mutation operator in this practice. Once enough offspring has been generated, the offspring is combined with their parents to form the next generation. The evolution process goes on until no significant changes

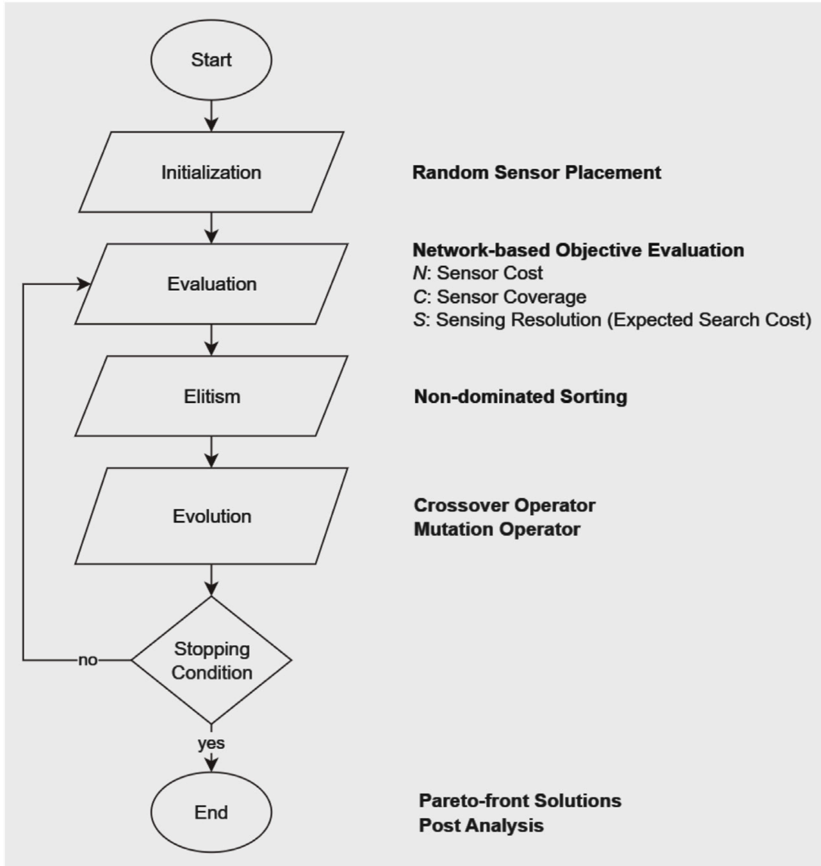


Fig. 3. A flowchart of the NSGA-II algorithm (Deb et al., 2002) coupled with network-based objective evaluation.

Table 1
Human heuristics for optimal sensor placement.

Ranking criteria	Definition
Indegree/outdegree (Leskovec et al., 2007)	The number of edges pointing inward to/outward from the node
Upstream size (Leskovec et al., 2007)	The number of nodes that can reach the node
Closeness centrality (Giudicianni et al., 2020)	The mean distance from a node to other nodes
Betweenness centrality (Giudicianni et al., 2020)	The extent to which a node lies on paths between other nodes
Katz centrality (Giudicianni et al., 2020)	The relative Influence of a node by considering the number of immediate neighbors and all other nodes that are connected with the node through this immediate neighbors

in objective values can be observed for 50 consecutive iterations. First-front plans in the final generation are regarded as Pareto-optimal sensor placement plans. Please refer to Deb et al. (2002) for more details of the standard NSGA-II algorithm.

Human heuristics. Rather than relying on optimization algorithms, human planners often follow different criteria to select sensor placement sites, which are mostly based on domain knowledge and practical experiences with implicit optimality. For example, Giudicianni et al. (2020) used five different network centrality metrics to select important manholes. Leskovec et al. (2007) proposed selection heuristics based on the number of in- or out-degrees. Based on existing literature, we selected a list of human placement heuristics (Table 1). The heuristics rank the importance of each manhole based on different aspects of network typologies. For example, betweenness centrality measures the importance of each manhole in constructing the shortest path for any other manhole pairs. Lastly, sensor placement plans based on human heuristics were simulated and compared with plans optimized by machine intelligence.

3.4. Model evaluation and performance analysis

We use the hypervolume indicator (HV) (Guerreiro, Fonseca, & Paquete, 2021) to evaluate the optimality of solutions throughout the optimization process and compare the quality of the Pareto-optimal solutions produced by different methods. HV measures the volume of the n-dimensional region between the solution set S and a reference point r. Due to the varying scales of different objectives, a max–min normalization process was conducted to transform all objective values to the range of 0 to 1 before the HV calculation. Therefore, in a 3-dimensional minimization problem, the reference point r was set to (1, 1, 1).

We further use two indicators, average monitoring ability (AMA, Eq. (6)) and the standard deviation of monitoring ability (STMA, Eq. (7)), to evaluate the sensor efficiency of each placement plan. While AMA measures the average sensor efficiency, STMA measures the variation of sensor efficiencies of all sensors in a plan. For an interconnected network component i composed by p sensors, AMA and STMA are formulated as follows,

$$AMA_i = \frac{\sum_{j=1}^{p} n_j}{p} \tag{6}$$

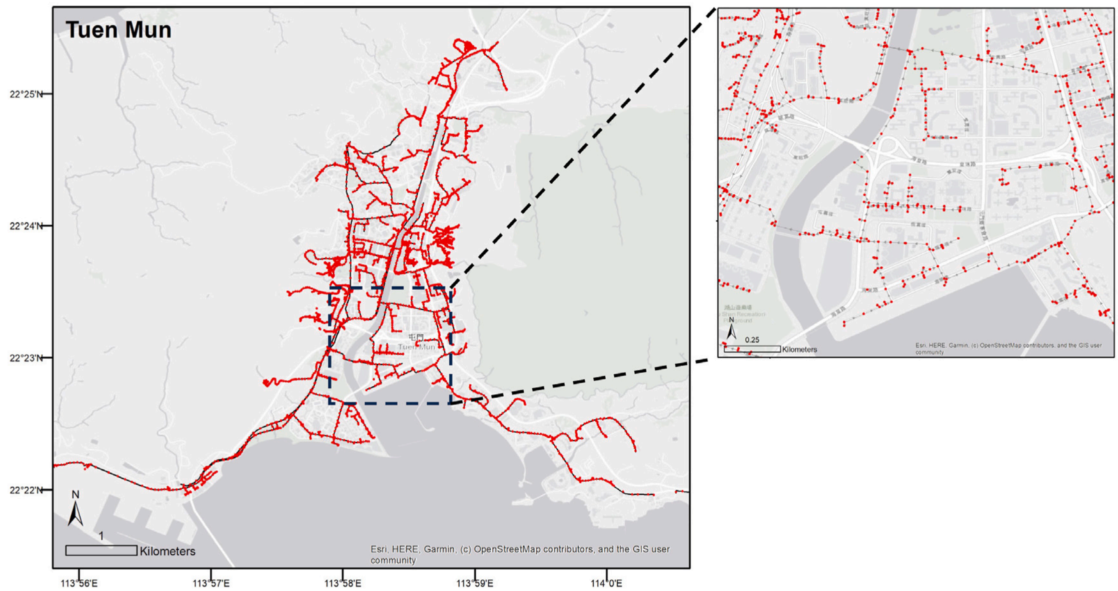


Fig. 4. Tuen Mun, Hong Kong, the study area sheltering more than 500,000 citizens, and its underground sewage network having 4,394 nodes (highlighted in red) and 4,308 edges.

$$STMA_i = \sqrt{\frac{\sum_{j=1}^{p}(n_j - AMA_i)^2}{p}} \tag{7}$$

where n_j is the size of the exclusive coverage of sensor j.

3.5. Study area and data

We demonstrate the effectiveness of the proposed model on an intricate sewage network located in Tuen Mun, Hong Kong including 4,394 nodes and 4,308 edges (Fig. 4). Tuen Mun is a high-density residential district in the northwestern part of Hong Kong's New Territories. With a population of more than 500,000, it is a bustling urban center with a diverse economy that includes manufacturing, logistics, and retail industries. The dataset was obtained from the Hong Kong GeoInfo Map platform (https://www.map.gov.hk/gm/). Necessary data preprocessing was executed manually to rectify any network topological issues.

4. Results

4.1. Optimization performance

Fig. 5a demonstrates the distribution of objective values of all initial and optimized sensor placement plans. Hyper-parameters were fine-tuned using a random search algorithm (Appendix A). Objective values of final-generation solutions fit well with the Pareto curve. We focus on solutions with more than 50% of the total coverage in the subsequent analyses.

The optimized solutions display an improvement in both quality and diversity compared to the initial generation (Fig. 5). The Pareto fronts move forward remarkably and the HV values increased from 0.249 in the initial population to 0.550 in the final generation, indicating a significant improvement in the quality of solutions. Optimized sensor placement plans have significantly higher sensor efficiency, with the average AMA increasing from 30.5 in the initial solutions to 41.5 in the final solutions, improved by 36.1%. Meanwhile, the average STMA converged slightly from the initial solutions (13.0) to the final solutions (12.9), indicating that the optimized placement plan slightly promotes equal sensor coverages. Average objective values are consistently improved over varying sensing costs (Fig. 5b). Interestingly, the optimization algorithm is more effective in improving sensing coverage when the sensor cost is small, while the algorithm is more effective in reducing the expected search cost when the sensor cost is large.

Trade-offs among sensing cost, sensing coverage, and expected search cost are demonstrated by the Pareto curves (Fig. 5a). As sensing costs increase, there is a corresponding increase in sensing coverages, coupled with a decrease in expected search costs for both initial and final solutions. Additionally, with more sensors deployed, the marginal effect of each sensor decreases. For example, 20 sensors can already provide an 80% coverage of all manholes, yet 80 sensors are required to achieve a 90% coverage.

In addition, the algorithm outputs a diverse generation of 200 optimal sensor placement plans with the sensing cost ranging from 2 to 100 and the expected search cost ranging from 5.2 to 10.4. Human decision-makers can further select the most feasible sensor placement plan from the pool of optimized plans. These results underscore the optimization algorithm's effectiveness in generating well-converged and diverse sensor placement plans.

We notice that a small number of *preferred manholes* are repeatedly selected for sensor placement. Fig. 6b summarizes the frequency distribution of the selected times for each manhole selected at least once by Pareto optimal solutions. Only 236 *preferred manholes* (5.3%) have been selected at least once, among which 25 manholes were selected more than 80 times, and only 2 manholes were selected more than 100 times. Notably, the number of manholes selected only once or twice is 59, higher than the number of manholes selected more times. Fig. 6a demonstrates the locations of the *preferred manholes*.

4.2. Examples of optimized sensor placement plans

Fig. 7 demonstrates two examples of the final solutions and one initial solution as a benchmark. While final solution 1 achieved the maximum sensor cost (100), final solution 2 achieved the maximum sensing coverage of 4018 manholes. The initial sensor placement plan covers 2835 manholes (64.5%) using 100 sensors with an expected search cost of 6.44. Despite that sensors cluster in areas having a high density of manholes, many manholes in areas A, B, C, and D are left uncovered. Using the same sensing cost, final solution 1 covers 3557 manholes (80.9%) with an expected search cost of 5.86. Final solution 2 covers 4018 manholes (91.4%) are covered using 86 sensors with an

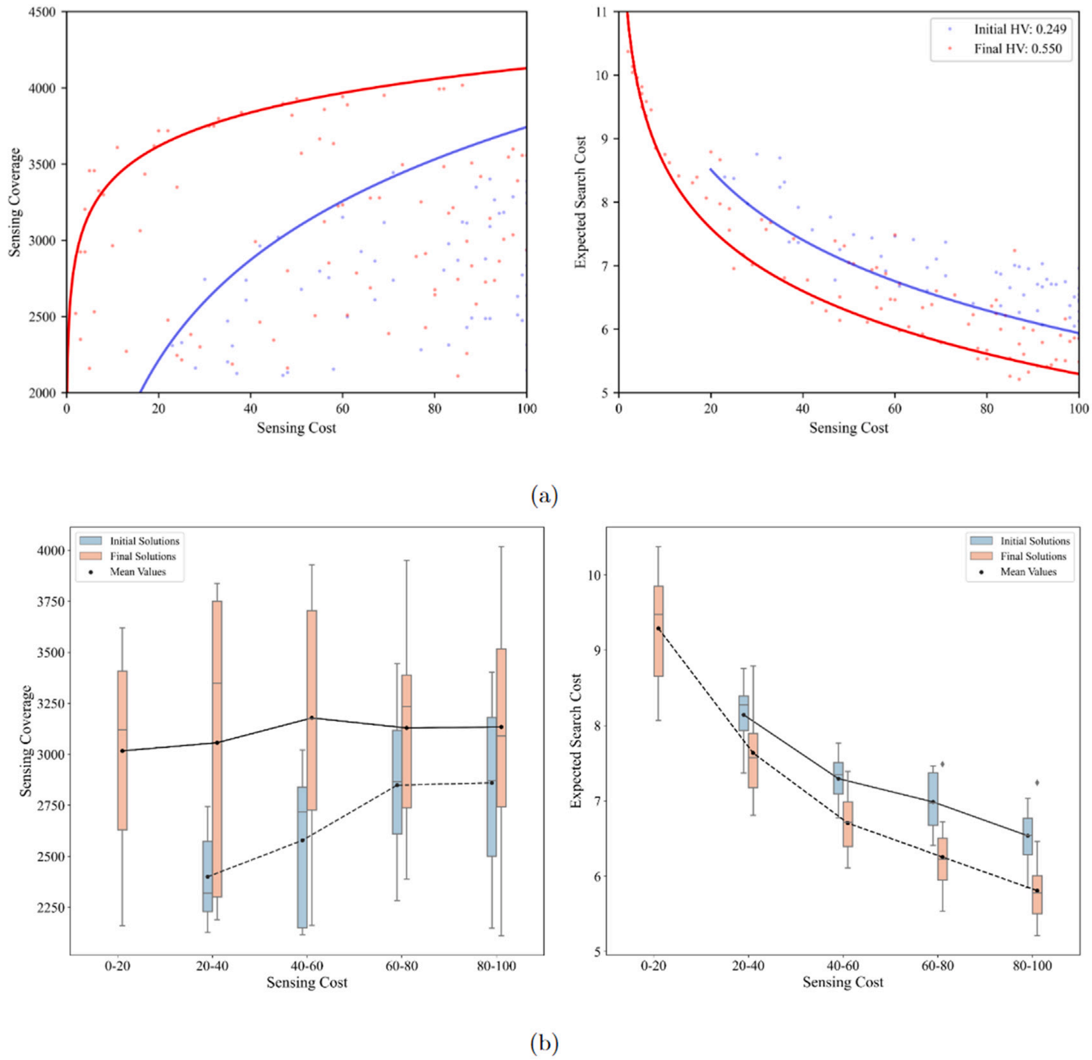


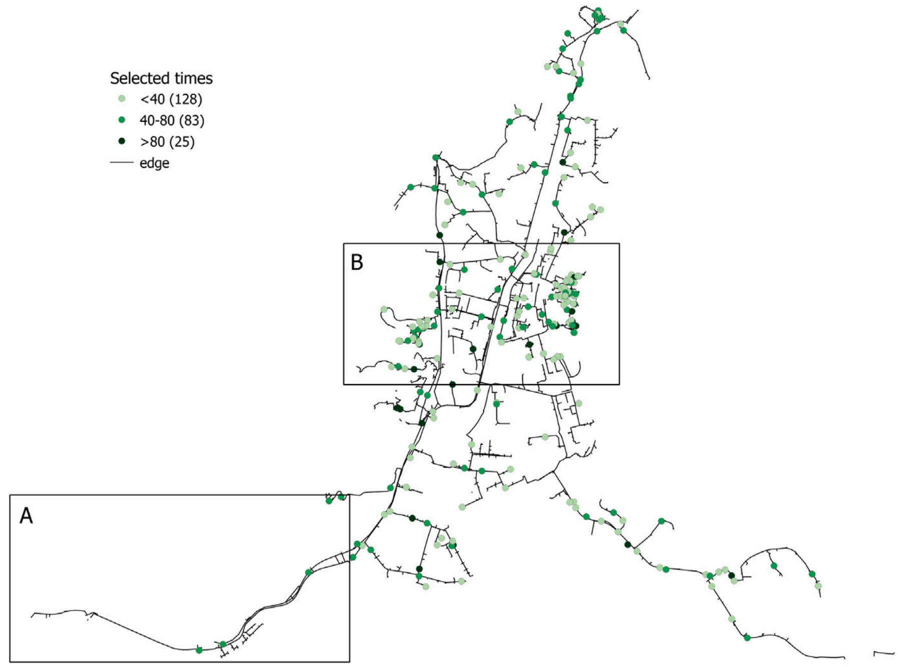
Fig. 5. Objective value distributions of initial and optimized sensor placement plans. (a) Paired Pareto fronts respectively between sensing cost and sensing coverage (left) and between sensing cost and expected search cost (right). (b) Average sensing coverage (left) and expected search cost (right) over varying sensing costs in initial and optimized sensor placement plans.

expected search cost of 7.24. Both final solutions achieved better cost-effectiveness in sensing. Compared to final solution 1, final solution 2 achieved better coverage using fewer sensors at the cost of a higher expected search cost. The comparison between final solutions 1 and 2 highlights the trade-off among Pareto optimal solutions.

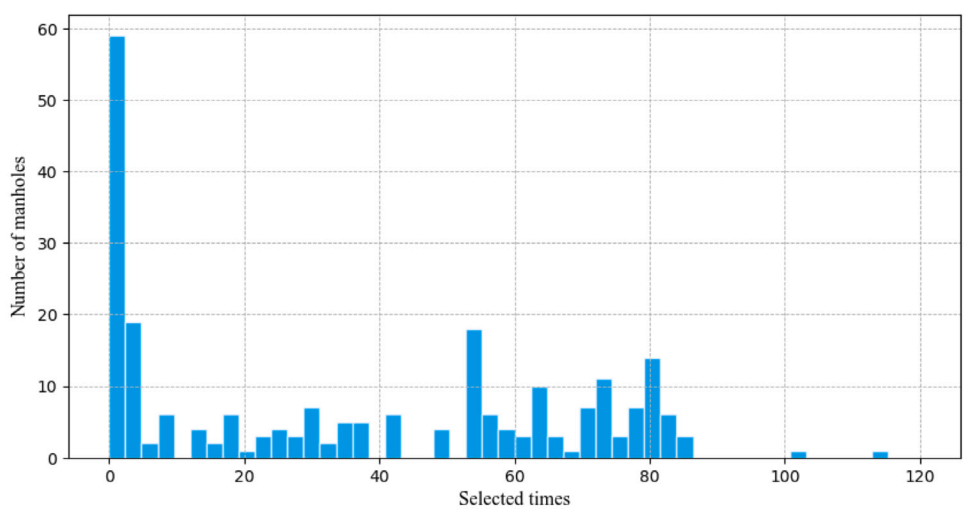
4.3. Comparison between machine intelligence and human heuristics

We compare the feasible solutions obtained by the MOSP model and human heuristics to gain insights into the difference between machine and human intelligence. Table 2 shows the normalized HV values. NSGA-II performs significantly better than all human heuristics, with the highest HV value of 0.550 and 102% larger than the best human heuristic based on outdegree. Random selection ranks the second with an HV of 0.194. The upstream size-based and closeness centrality-based human heuristics perform worse than random selection. In contrast, indegree, betweenness centrality, and Katz-centrality-based human heuristics failed to generate any feasible solution covering more than half the total number of manholes. Furthermore, we investigate the relationship between algorithm performance and sensing cost (Fig. 8). All algorithms' HV values increase with higher sensing costs, while NSGA-II consistently outperforms other heuristics by a significant margin, demonstrating its superior performance in generating cost-effective sensor placement plans.

Fig. 8b demonstrates the paired Pareto front and the number of feasible solutions, where NSGA-II retains 90 solutions, followed by three human heuristics. Most solutions in the Pareto front of NSGA-II dominate the solutions obtained by human heuristics. The ranges of objectives obtained by NSGA-II are greatly larger than those of human heuristics. These results indicate higher quality and richer diversity of the solutions obtained by NSGA-II. As mentioned by Leskovec et al. (2007), outdegree indicates the node's tendency to summarize the information. Considering a sensor A placed in the manhole with a large outdegree in our problem, if A gives an alarm, then all the downstream manholes need not be tested, which can help save lots of sampling costs. If A does not give an alarm while the downstream sensor B indicates the appearance of the virus, then the sampling cost can also be reduced greatly since only one branch between A and B needs to be tested. Therefore, selecting sensor locations based on outdegree can perform best among heuristics because these manholes are an excellent summary of the sewage network despite their unpopularity in practice. For the upstream size and closeness centrality-based human heuristic, no obvious improvement in sensing coverage can be observed



(a)



(b)

Fig. 6. The spatial (a) and frequency distribution (b) of the selected times for *preferred manholes*. Only manholes selected at least once by Pareto optimal solutions are included.

as the sensing cost increases. Turning to the feature map shown in Appendix B, we notice that the manholes with high upstream size and closeness centrality values are similar. Intuitively, placing one sensor in these manholes contributes little to the overall performance. This observation also holds for the betweenness centrality-based heuristic. However, such distribution does not exist for indegree, outdegree, and Katz centrality.

Our analysis reveals that human heuristics do not perform as well as machine intelligence. The chosen manholes based on topological characteristics tend to be close to each other, leading to sensors spread out too little. Besides, relying solely on intuition may overlook important manholes, such as those with large outdegrees. By contrast, machine intelligence can overcome these tendencies and generate a more reasonable and effective placement plan. Our results suggest that machine intelligence can significantly enhance the efficiency and effectiveness of sensor placement in sewage networks, ultimately leading to better monitoring and control of sewage systems.

5. Discussion

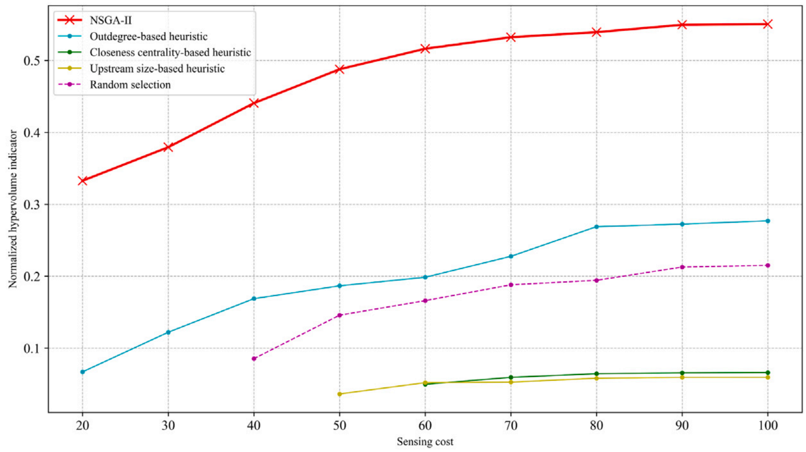
Developing a cost-effective monitoring and warning system on city-scale sewage networks is vital for protecting urban public health during pandemics. Effective detection of SARS-CoV-2 viral genome from sewage water has been reported in multiple global cities such as Brisbane, Australia (Ahmed et al., 2020), Jaipur, India (Arora et al., 2020), and Hong Kong (Deng et al., 2022), yet insufficient attention has been paid to the sensor placement strategy of city-scale sewage surveillance systems. For example, the Brisbane study (Ahmed et al.,

(a) final solution 1 (maximum sensing cost)

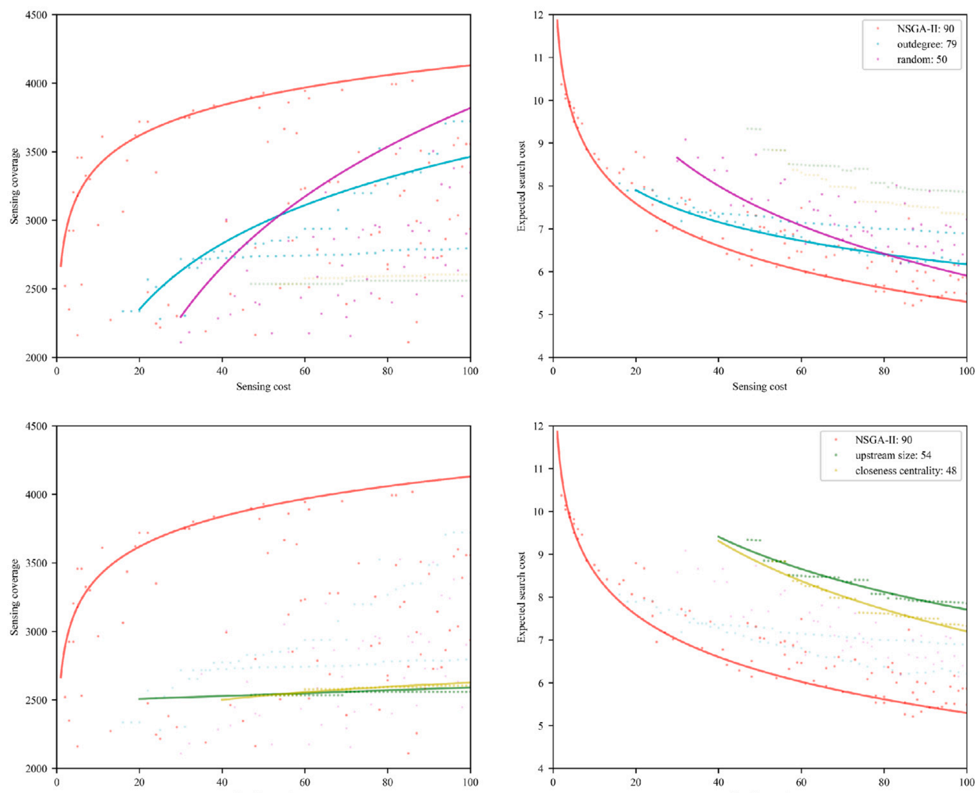
(b) final solution 2 (maximum sensing coverage)

Fig. 7. Two examples of optimized sensor placement plans and an initial plan as a benchmark. Rectangle (A)-(J) mark uncovered areas.

2020) collected samples from one suburban pumping station and two municipal wastewater treatment plants (WWTPs), demonstrating accurate prevalence estimations of the SARS-CoV-2 viral genome among over 1 million urban residents using only three sampling sites. The Jaipur study (Arora et al., 2020) collected sewage samples from six WWTPs and two hospitals and reported correlations between viral genome signals and public health data. Hong Kong implemented one of the world's earliest attempts at large-scale sewage surveillance to monitor the epidemic's progression in communities, adopting 492 sampling sites to cover its seven million population (Deng et al., 2022). The sampling sites were manually selected based on government-specified heuristics. A higher density of sampling sites provides fine-grained



(a)



(b)

Fig. 8. Performance comparison between optimization and human heuristics. (a) Normalized HV values for NSGA-II and human heuristics over varying sensing costs. (b) Objective value distributions and paired Pareto fronts respectively between sensing cost and sensing coverage (left) and between sensing cost and expected search cost (right).Each scatter point represents a solution in the objective space.

information on viral prevalence and enables early outbreak detection in communities. A strategic sensor placement method becomes necessary with an increased number of sampling sites.

We proposed a multi-objective optimization framework for cost-effective sensor placement at a large scale. Larson et al. (2020) propose a Bayesian-probability-based approach assuming that a sewage network follows a single-origin tree structure, which is not always true in reality. Our method stands out by its generalizability to all kinds of network structures. We integrated the Bayesian probability-based approach to measure the expected search cost of individual sensors, making the proposed method applicable to all sewage networks. Studies in Girona, Spain (Calle et al., 2021) and Suwon, Republic of Korea (Kim et al., 2022) respectively proposed sensor placement methods based on an improved greedy algorithm and a single-objective genetic algorithm. Both studies demonstrated limited scalability and were tested in problems having up to 16 sampling sites. The limited scalability was mainly due to the use of a coverage matrix representing whether the monitoring station covers a node, requiring $O(N^2)$ memory space for N manholes. As the network grows larger, the coverage matrix becomes more sparse, making the calculation slow and inefficient. In contrast, our method explicitly considers upstream-downstream relationships in sewage networks by constructing a connectivity subgraph. Objective assessments

Table 2
Normalized hypervolume indicator for NSGA-II and human heuristics.

Methods	Normalized hypervolume indicator
NSGA-II	0.550
Outdegree-based heuristic	0.271 (0.018)
Random selection	0.194 (0.018)
Closeness-centrality-based heuristic	0.066 (0)
Upstream-size-based heuristic	0.059 (0)
Indegree-based heuristic	[a]
Betweenness-centrality-based heuristic	[a]
Katz-centrality-based heuristic	[a]

Random selection was conducted among maholes having identical indegree, outdegree, upsteam size, and closeness centrality values. The experiment was repeated 10 times to report HV standard deviations in brackets.

[a] Could not achieve the 50% minimal coverage constraint.

using the connectivity subgraph reduce the space complexity to $O(n)$, where n is the number of sensors and is usually much smaller than N. By integrating topological characteristics of directed sewage networks in objective assessments, the proposed optimization algorithm demonstrated superior efficiency and scalability, making it more practical for real-world, city-scale sewage network surveillance applications.

The proposed method uses computational intelligence to generate cost-effective sensor placement plans in large-scale sewage networks. The method demonstrated significantly better performance compared to traditional human sampling strategies based on simple heuristics. Taking degree-related heuristics as an example, people may prefer manholes with larger indegrees to those with large outdegrees because larger indegrees allow a sensor to cover more manholes. Counterintuitively, our results show that selecting manholes with large outdegree performs better.

Limitations exist in this paper and could be improved in future research. Meta-heuristic optimization methods, such as NSGA-II, do not provide an assurance of achieving global optimality in the results obtained. Conducting performance comparisons with results obtained from alternative algorithms can aid in uncovering globally optimal solutions for this problem. Literature has reported that the traceability of the virus may decrease as sewage water travels downstream. Coupling a simulation model can contribute to a more accurate model. Manholes may have varying accessibilities due to rapid urban changes. In practice, field inspections are required to verify manhole-specific feasibility for sewage water sampling. Lastly, future research should focus on obtaining the necessary resources to conduct field experiments for a more robust evaluation of the method's efficacy. These limitations provide opportunities for future research.

6. Conclusion

Large-scale urban sewage surveillance is becoming a fundamental urban sensing infrastructure in cities. Sewage surveillence generates vital information for extensive urban applications, ranging from early outbreak detection in a pandemic to post-pandemic, long-term public health monitoring tasks such as antibiotic-resistant virus monitoring and community drug use detection. This paper proposes an network-based optimziation approach to enable cost-effective sensor placement in large-scale sewage networks. The proposed optimal sensor placement (OSP) model simultaneously maximizes the sensing effectiveness (*i.e.*, sensing coverage andexpected search cost) and minimizes the sensing cost. The model incorporates an evolutionary optimization algorithm (NSGA-II) embedded with a connectivity-based objective evaluation approach to enable efficient optimization on large-scale directed networks. A case study in Tuen Mun, Hong Kong shows that the optimization model significantly outperforms human placement heuristics in generating high-quality sensor placement plans, highlighting the importance of machine intelligence in supporting cost-effective urban sensing.

CRediT authorship contribution statement

Sunyu Wang: Methodology, Optimization, Data Analysis, Manuscript Preparation. **Ke Xu:** Optimization, Data Analysis. **Yulun Zhou:** Conceptualization, Problem Formulation, Methodology, Project Supervision, Manuscript Preparation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships.

Data availability

The codes and datasets used in this study are openly available upon request at https://github.com/hkuzebralab/spo-nsga.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.scs.2024.105250.

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